

Aether One™ – Resource Snapshot v1

Assessment of human, computational, and financial resources for Aether One RISC-V SoC and prototype development.

1. Human Resources & Expertise

Area	Current Status	Needs / Gaps	Action Notes
RISC-V / RVV Development	Good familiarity with toolchain and architecture with no real hands-on with the RVV hardware (HiFive or StarFive board)	Develop cross-compiler toolchain, order development board	
AI Inference / ONNX Runtime	Strong understanding at conceptual level, simulation on RISC-V native environment	Test ONNX Runtime Lite or TVM on RVV-enabled hardware	
Security / PQC	Initial familiarity with liboqs & TPM concepts	Integration and testing of PQC libraries (e.g. liboqs) on RISC-V	
Sensor Fusion / ROS-PCL	ROS/PCL stack awareness, data-level integration ready	Simulate LiDAR + BLE + spectral data fusion on RISC-V	
Low-Level OS / Kernel	Good Linux / RTOS background	Custom microkernel profiling and configuration	
		Build and integrate AetherOS RT microkernel image	

2. Compute & Hardware Resources

Component	Current Capability	Needed Upgrade / Add-on	Action Notes
Raspberry Pi 5 Stack	Functional for software validation	Limited RVV / RISC-V emulation only	Continue as base dev environment
RISC-V Dev Board	Not yet available	HiFive Unmatched / StarFive VisionFive A2 (FPGA or RISC-V)	Acquire FPGA (Pilot) for performance tests
Sensors	BLE, LiDAR, VOC, Geiger modules readily available	Integration and synchronized timestamping	Test Sensor Fusion Bus on Pi 5
Networking / Cloud	MQTT/REST/WS stable	Quantum Connector API integration	Prototype Debian Quantum AI bridge
Workstation / Toolchain	Python, GCC/LLVM, Docker configured	RISC-V cross-compiler with PQC extensions	Build and benchmark toolchain

3. Budget & Timeline Overview

Phase	Estimated Cost (€)	Duration	Notes / Actions
Prototype Validation (Pi 5)	≈ 500	1–2 months	Low-cost development, focus on software and sensor integration
RISC-V Hardware Evaluation	500–1500	2–4 months	Purchase dev board or FPGA; validate RVV performance
ASIC Design Simulation	≈ 25,000	3–6 months	Optional: engage VTT or university partner
Engineering Sample (ASIC)	≈ 250,000+	12–18 months	Requires consortium / investor funding
Deployment & Scaling	TBD	Post-2026	Dependent on prototype results

Summary:

The current resources are sufficient for continued prototype development on Raspberry Pi 5 and RISC-V software simulation. Next-phase readiness depends primarily on obtaining an RVV-capable development board and allocating modest funds for hardware testing. The human resource base is strong; expansion is required only for ASIC-level design and production.